

Investigation on pathogenic microorganisms in poultry chicken meat from local markets of Rajshahi, Bangladesh

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ABSTRACT

Around one quarter of the dietary protein in chicken meat makes it a very economical and nutrient-dense source of protein. Research indicates that 37% of Bangladesh's gross production of meat comes from chicken. In keeping with prior investigations, raw meat is vulnerable to a range of microorganisms, especially bacteria that may infest people with food-borne illnesses and trigger abdominal pain, nausea, and diarrhea. That is why we sought to find out if there were indeed potentially pathogenic bacteria in raw meat of broiler, layer and sonali varieties of poultry. We collected meat specimens from Binodpur and Shaheb Bazaar and then isolated and purified microorganisms from mixed cultures. Afterwards, the characterization of isolated bacteria was done by different morphological and biochemical test methods. Based on those assays, the isolates were identified as *Klebsiella*, *Salmonella*, *Staphylococcus*, *Shigella*, and *Escherichia* species. All these bacteria are potentially pathogenic to human. Additionally, tests for antibiotic susceptibility were performed, and it was discovered that, with the exception of imipenem, Ciprofloxacin, and Rifampicin susceptibility, all isolates displayed resistance to various antibiotics, indicating multidrug resistance. The current study shows that multiple antibiotic-resistant pathogenic bacteria can be encountered in chicken meat and constitute a serious threat to the public health. Given that microbial contaminants have been discovered in raw meat all over the world including Bangladesh, we can recommend people be extra cautious when handling raw meat.

INTRODUCTION

Poultry chicken, comprising various breeds, denotes domesticated birds raised for meat and eggs. It holds global agricultural and economic importance, providing vital protein and livelihoods. Bangladesh has seen a surge in poultry consumption due to rising incomes and diverse protein demands¹. Consequently, the poultry industry plays a key role in meeting the nation's animal protein needs.

However, this production upswing raises concerns about potential transmission of pathogenic microorganisms through poultry products, posing risks to human health during handling, processing, and cooking. Previous studies confirm the presence of various microorganisms like *E. coli*, *Salmonella* spp., *Campylobacter* spp., *Clostridium* spp., *Staphylococcus* spp., *Klebsiella* spp., and *Shigella* spp. in raw poultry meat, leading to foodborne diseases. Contamination rates parallel the rise in poultry production.

A study in Bangladesh found that 98% of commercial chicken farms use antimicrobials in production, with 85% using them prophylactically². Antimicrobial resistance among foodborne pathogens has surged in recent decades. This study addresses the critical issue by examining pathogenic microorganisms in poultry chicken meat from local Rajshahi markets.

Methodology

Broiler chicken
Sonali chicken
Layer chicken

Isolation and purification of bacteria from chickens raw meat

Characterization of isolated bacteria

Morphological tests
1. Gram Staining
2. Motility Test

Biochemical tests
1. MacConkey Agar Test
2. Mannitol Salt Agar Test
3. Bismuth Sulphite Agar Test
4. Xylose Lysine Deoxycholate Agar Test
5. E.M.B Agar Test
6. Starch Test
7. Urease Test
8. Starch Hydrolysis Test
9. Simmons Citrate Agar Test
10. Catalase Test
11. Methyl Red Test
12. Blood Agar (Haemolytic) Test

Antibiotic Sensitivity test
Antibiotic Sensitivity Test

RESULTS & DISCUSSION

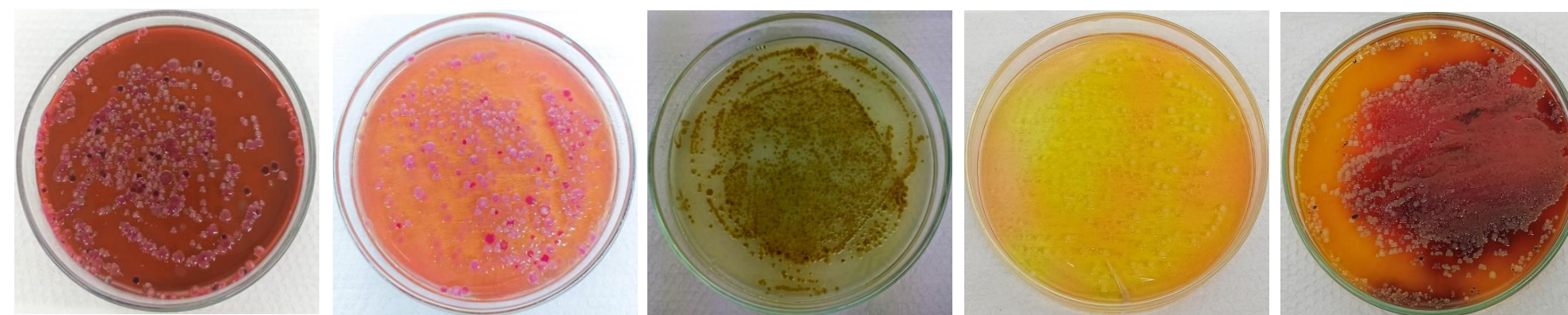


Figure: Mixed culture from meat samples

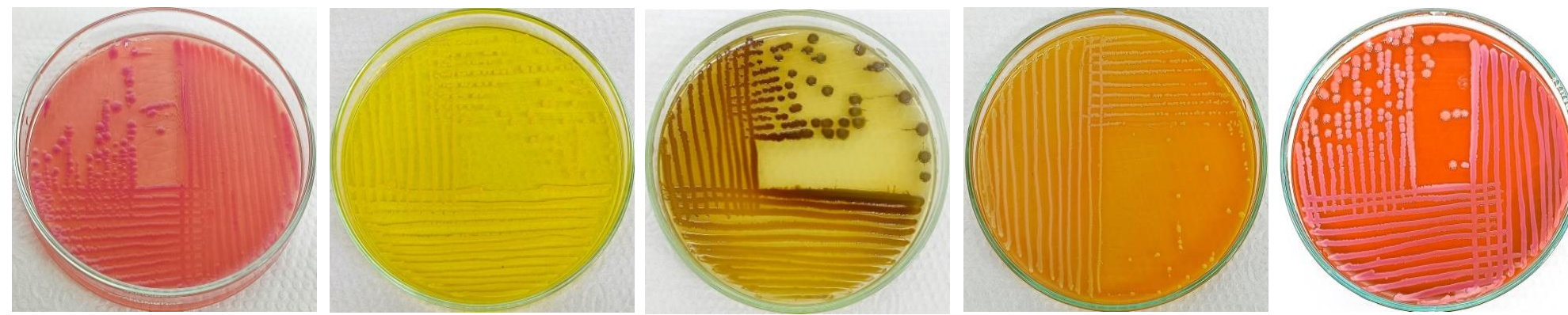
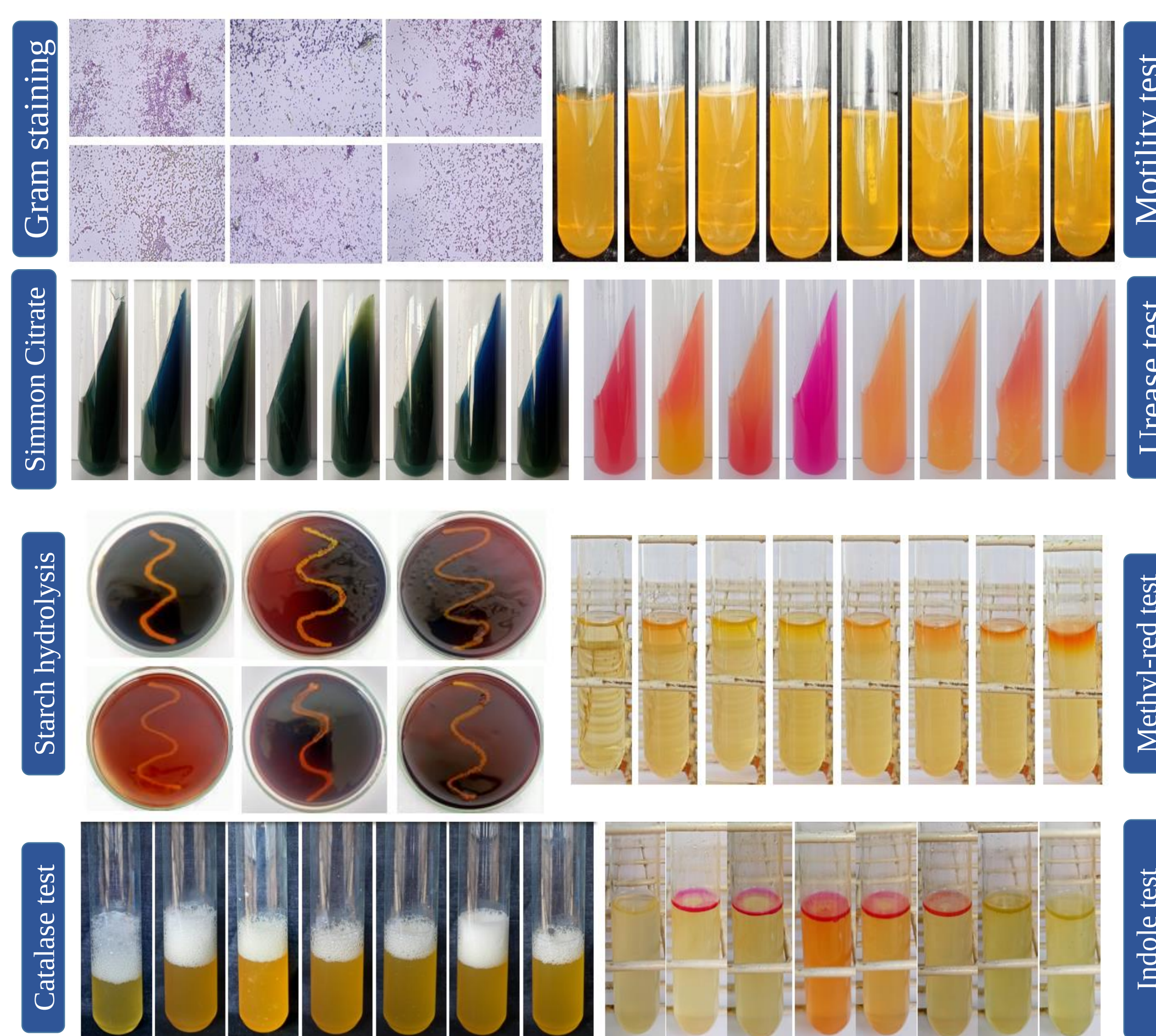


Figure: Pure culture isolation on selective media

Morphological & Biochemical tests



Summary of Morphological & Biochemical test results

Test name	Results				
	Iso-1a	Iso-1b	Iso-2	Iso-3	Iso-4
MacConkey	-	-	-	-	-
E.M.B	-	-	-	-	-
BSA	-	-	-	-	-
Simmons citrate	-	-	-	-	-
Starch hydrolysis	+	+	+	+	+
Urease	+	+	+	+	+
MSA	+	+	+	+	+
T.S.I. agar	+	+	+	+	+
Motility	+	+	+	+	+
Catalase	+	+	+	+	+

Broiler chicken

Sonali chicken

Layer chicken

Test name	Results					
	Iso-1a	Iso-1b	Iso-1c	Iso-1d	Iso-2a	Iso-2b
Gram staining	-	-	-	-	+	+
MacConkey	+	+	+	+	-	-
E.M.B	-	+	-	-	+	+
BSA	-	-	-	-	-	-
Simmons citrate	+	+	+	+	-	-
Starch Hydrolysis	-	+	-	-	-	-
Urease	-	+	-	-	+	+
MSA	-	-	+	+	+	+
T.S.I.	+	+	+	+	+	+
Motility	+	+	-	-	+	+
Catalase	+	-	+	+	+	+

Test name	Results						
	Iso-1	Iso-2	Iso-3	Iso-4	Iso-5	Iso-6	Iso-7
Gram Staining	-	+	-	-	-	-	-
Motility Test	+	+	-	-	+	+	+
MacConkey	+	-	-	-	-	-	-
MSA	-	+	-	-	-	-	-
BSA	-	-	+	-	-	-	-
XLD	-	-	-	+	-	-	-
EMB	-	-	-	-	+	+	+
Starch Hydrolysis	-	-	-	-	-	-	-
Simmons Citrate	+	-	-	+	-	+	+
Urease	+	+	+	-	-	+	+
Catalase	+	+	+	+	+	+	+
Methyl Red	+	-	+	+	+	+	+
Blood Agar	-	-	-	-	-	-	-
Indole	+	+	+	+	+	-	-

ANTIBIOTIC SENSITIVITY TEST RESULTS

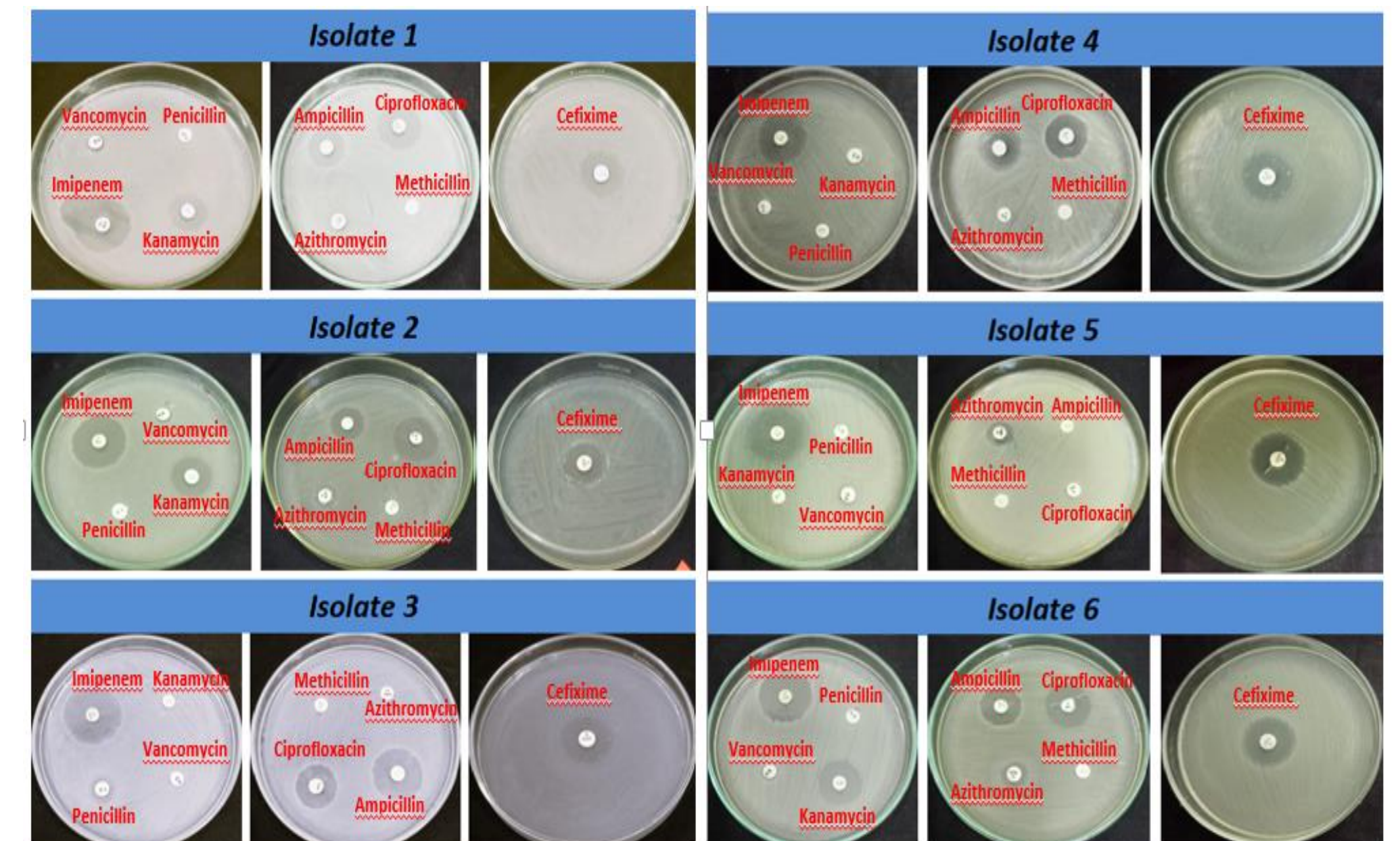


Figure: Antibiotic sensitivity test of different bacterial isolates.

Table: Antibiotic sensitivity result of bacterial isolates from Broiler chicken

Name of Antibiotic	Zone of inhibition (mm)				
	Isolates				
	1a	1b	2	3	4
Imipenem	36	37	30	33	52
Penicillin	8	24	6	23	38
Aztreonam	6	8	6	6	8
Cefixime	11	13	6	10	16
Vancomycin	15	14	17	15	22
Kanamycin	15	14	18	13	12
Ciprofloxacin	10	16	11	6	11
Ampicillin	6	18	6	14	21
Azithromycin	6	7	6	6	6
Methicillin	22	20	18	21	22
Doxycycline	6	6	6	6	6
Rifampicin	22	22	20	18	21

Broiler chicken

Table: Antibiotic sensitivity result of bacterial isolates from Sonali chicken

Name of Antibiotic	Zone of inhibition (mm)					
	Isolates					
	1a	1b	1c	1d	2a	2b
Imipenem	15	15.5	22.5	21.5	36	36
Penicillin	6	7	6	7	32	34.5
Aztreonam	19.5	17.5	23.5	25.5	9	10
Cefixime	6	7	6	6	18	18
Vancomycin	7.5	9.5	7.5	7.5	21	21
Kanamycin	16	10.5	16.5	15	22	22
Ampicillin	6	7	6	6	25	25
Azithromycin	8.5	15	8.5	8.5	6	6
Methicillin	6	8	7	6	26	26
Ciprofloxacin	28	27	28	27	26	26
Cefuroxime	14	9	14	9	18	18

Sonali chicken

Table: Antibiotic sensitivity result of bacterial isolates from Layer chicken

Name of Antibiotics	Zone of inhibition (mm)						
	Isolates						
	1	2	3	4	5	6	7
Kanamycin	17.5	18.5	6	6	6	20	6
Vancomycin	7	7	6	6	7	6	7
Imipenem	28	26	26.5	26	30	26.5	31.5
Ampicillin	19.5	18	23	19.5	6	20	13
Methicillin	6	6	6	6	6	6	6
Azithromycin	7	6	6	6	12	12	11.5
Cefixime	19	20	36	20	22	12	26.5
Penicillin	6	7	6	6	6	6	6
Ciprofloxacin	22.5	23	19.5	19.5	6	22	12.5

Note-1: Resistant<10 mm; Intermediate resistant =10-15 mm; Susceptible>15 mm. (Kirby and Bauer, 1996) Note-2: R= Resistant, IR= Intermediate Resistant, S= Susceptible

SUMMARY

- ❖ The isolated bacterial strains might be *Klebsiella*, *Salmonella*, *Staphylococcus*, *Shigella*, and *Escherichia* species.
- ❖ All the isolated bacteria are potentially pathogenic.
- ❖ All of the isolated bacteria were observed to be multidrug resistant.
- ❖ The fact that, we have imipenem, Rifampicin, and ciprofloxacin to treat them, on the other hand, gives us hope.
- ❖ As pathogenic bacteria have been identified from raw layer meat all across the world, including in Bangladesh, we can encourage people to be more concerned regarding processing, storing and cooking poultry chicken meat.

REFERENCES

- Da Silva, C.A. and M. Rankin, 2014. Contract Farming for Inclusive Market Access. Food and Agriculture Organization of the United Nations, Rome, Italy, ISBN: ISBN 978-92-5-108061-0, Pages: 227.
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